Find your method

Open the book here when you don't know which chapter to read. Three routes are offered: a flowchart from research question to statistical test, a quick lookup table, and a click-through wizard hosted live.

- **Interactive wizard** — three to five questions, one recommended test.
- **Static decision tree** (below) — the same logic, single page, suitable for printing.
- **Quick lookup table** — common scenarios mapped to a recommended chapter and section.

This chapter covers the *core* tree of classical tests (differences, associations, interdependence). Methods outside that core — survival, Bayesian, machine learning, time series, multivariate, meta-analysis, bioinformatics, study-design choices — are reached through their topic chapters or via the lookup table below.

The static decision tree

Colour encodes the scale level of the outcome; a dashed border marks a non-parametric test. Click a node in the rendered HTML to jump to the method page.

```
%%{init: {"securityLevel": "loose", "flowchart": {"htmlLabels": true, "useMaxWidth": true}}}%%
flowchart TD
    START["Research question"]:::root --> Q2{"Differences or<br/>associations?"}
    START --> Q_INT{"Exploratory<br/>structure?"}

    Q2 -->|Differences| QD1{"Kind of difference?"}
    Q2 -->|Associations| QA1{"Variable count?"}

    QD1 -->|Central tendency| QD2{"Outcome scale?"}
    QD1 -->|Variances| VAR["Variance tests<br/>(chi-sq, F, Levene)"]:::metric
    QD1 -->|Proportions| QDPROP{"Single variable vs. expected?"}

    QDPROP -->|Dichotomous| BIN["Binomial test"]:::nominal
    QDPROP -->|Categorical| CHI_GOF["Chi-sq goodness-of-fit"]:::nominal

    QD2 -->|Interval/ratio| QD3{"Number of groups?"}
    QD2 -->|Ordinal| QD6{"Groups &amp; pairing?"}
    QD2 -->|Nominal| QDPROP

    QD3 -->|2 groups| QD4{"Independent/paired?"}
    QD3 -->|3+ groups| QD5{"Factors &amp; design?"}

    QD4 -->|Indep. normal| TT["Independent t-test"]:::metric
    QD4 -->|Indep. non-normal| MWU["Mann-Whitney U"]:::metric_np
    QD4 -->|Paired normal| PT["Paired t-test"]:::metric
    QD4 -->|Paired non-normal| WSR["Wilcoxon signed-rank"]:::metric_np
    QD4 -->|Paired sign only| SGN["Sign test"]:::metric_np

    QD5 -->|1 between normal| OWA["One-way ANOVA"]:::metric
    QD5 -->|2+ between normal| FA["Factorial ANOVA"]:::metric
    QD5 -->|1 within normal| RM1["One-way rmANOVA"]:::metric
    QD5 -->|Mixed design| RMM["Factorial/mixed rmANOVA"]:::metric
    QD5 -->|Indep. non-normal| KW["Kruskal-Wallis"]:::metric_np
    QD5 -->|Paired non-normal| FR["Friedman"]:::metric_np

    QD6 -->|2 indep.| MWU
    QD6 -->|2 paired| WSR
    QD6 -->|&gt;2 indep.| KW
```

```
    QD6 -->|&gt;2 paired| FR

    QA1 -->|Two variables| QA2{"Scale levels?"}
    QA1 -->|More than two| QA5{"DV scale?"}

    QA2 -->|Both nominal| CHIC["Chi-sq contingency"]:::nominal
    QA2 -->|Both ordinal| SPE["Spearman"]:::ordinal_np
    QA2 -->|Ord + interval| SPE
    QA2 -->|Interval linear| PEA["Pearson"]:::metric
    QA2 -->|Interval directed| SLR["Simple linear regression"]:::metric
    QA2 -->|Interval non-norm| KEN["Kendall tau"]:::metric_np

    QA5 -->|Interval| MLR["Multiple linear regression"]:::metric
    QA5 -->|Dichotomous| LOG["Logistic regression"]:::nominal
    QA5 -->|Ordinal| ORL["Ordinal logistic regression"]:::ordinal
    QA5 -->|Nominal k+| MNL["Multinomial logistic regression"]:::nominal

    Q_INT -->|Reduce variables| FAC["Factor analysis"]:::interdep
    Q_INT -->|Group cases| QCL{"Scale/size?"}

    QCL -->|Mixed/small| HCL["Hierarchical clustering"]:::interdep
    QCL -->|Metric/large| KMC["K-means"]:::interdep
    QCL -->|Mixed/xlarge| TSC["Two-step clustering"]:::interdep

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```

Quick lookup

A small table for the most common biomedical scenarios. The full tree above and the live wizard cover everything the table omits.

Scenario	Recommended method	Chapter
2 independent groups, metric, normal	Independent / Welch t-test	@ref(sec-ch-04-inferential)
2 independent groups, non-normal or ordinal	Mann-Whitney U	@ref(sec-ch-04-inferential)
2 paired groups, metric	Paired t-test	@ref(sec-ch-04-inferential)
3+ groups, between, normal	One-way ANOVA	@ref(sec-ch-04-inferential)
3+ groups, between, non-normal	Kruskal-Wallis	@ref(sec-ch-04-inferential)
Two continuous, linear	Pearson correlation	@ref(sec-ch-04-inferential)
Continuous outcome, 2+ predictors	Multiple linear regression	@ref(sec-ch-07-regression)
Binary outcome, 1+ predictors	Logistic regression	@ref(sec-ch-07-regression)
Time-to-event with censoring	Kaplan-Meier, Cox	@ref(sec-ch-11-survival)
Diagnostic test accuracy	Sensitivity/specificity, ROC	@ref(sec-ch-14-clinical)
Aggregating effect sizes across studies	Pairwise meta-analysis	@ref(sec-ch-15-metaanalysis)

Scenario	Recommended method	Chapter
Predictive model with cross-validation	tidymodels workflow	@ref(sec-ch-13-ml)
Posterior inference with priors	brms / Stan	@ref(sec-ch-10-bayesian)
Repeated measures within subjects	Mixed-effects model	@ref(sec-ch-16-design)
Two or more variances	Levene's test	@ref(sec-ch-04-inferential)
Reduce variables to latent dimensions	PCA / factor analysis	@ref(sec-ch-08-multivariate)

The static tree is mirrored from the decision-tree wizard at https://cttir.github.io/tutorials/decision-tree/. Structure inspired by the University of Zurich Methodenberatung (methodenberatung.uzh.ch).